Enrollment No. 0160117207

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:B.Tech., Semester:6th
(May, 2024)

Subject Code: BAI-304	**Subject: Deep Learning-II**
Time :3 Hours	**Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(5*4=20)**
	(a) Why is the sigmoid function not suitable for CNN models? Discuss the approaches to mitigate this.	
	(b) Analyze the relationship between regularization and vanishing gradient? Explain with proper Justification	
	(c) Discuss the relationship between autoencoder and PCA..	
	(d) Is the number of neurons in the output layer tuned through hyperparameter tuning? Which method do we follow for this?	
UNIT-I		
Q2	a) Consider a three-input perceptron with weight vector a [0.1,0.3,0.4] and bias of 0. For the input vector of [0.1, -0.4, -0.5] and output label as 1, what will be the output of perceptron with sigmoid, tanh, and ReLU activation function? Considering the outputs on these activation functions, suggest the suitable one and why? b) Can zero initialization of the NN weights with dropout lead to symmetry? Justify your answer	**(5+5=10)**
Q3	a) Why do we consider dimensionality reduction methods, like PCA, as unsupervised methods irrespective of whether the data is labeled or not? Explain. b) On an edge computing device with 1kb of memory, ANN model needs to be trained in the fastest possible time on dataset in which each record occupies 800 bytes. Which optimizer is best suitable and why?	**(5+5=10)**
UNIT-II		
Q4	Discuss the relationship between early stopping and dropout in context of training a deep learning model. Is it beneficial to use both in a DL model? Justify your answer.	**(10)**
Q5	Explain the working of Adagrad and momentum gradient decent optimizers in detail. How do they prevent the optimization process from getting stuck in local minima Also, which one is better than batch gradient descent? Justify your answer.	**(10)**
UNIT-III		
Q6	Consider the following values of confusion matrix: TP = 90, FP=10, FN=30, and TN=10. How many there are total positive and negative samples in the dataset. Also, calculate Accuracy, Precision, Recall, sensitivity, specificity, and F1-score for this.	**(10)**
Q7	a) Why do we have different performance metrics for the different problems? b) Explain the need for hyperparameter tuning in DL model. Differentiate between Grid and Random hyperparameter tuning.	**(5+5=10)**
UNIT-IV		
Q8	a) Explain the GAN with the architecture. Discuss the behavior of generative model in comparison to discriminator model. b) How does a convolutional autoencoder perform decoding? Explain in detail.	**(5+5=10)**
Q9	a) Explain the undercomplete autoencoder through diagram. What are the benefits of a regularized autoencoder over an undercomplete autoencoder? b) Discuss the loss functions of GAN architecture. How these loss functions lead to better training.	**(5+5=10)**